

HALIFAX NORTHAMPTON, NC, USA

WMO: 720649

Lat: 36.330N

Lon: 77.635W

Elev: 145

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00231

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	22.7	1.2	5.9	25.2	7.0	7.9	27.3	21.3	54.1	18.9	51.5	4.6	240	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	94.9	76.3	91.2	75.8	90.0	75.7	80.2	88.5	78.4	87.2	77.3	86.2	8.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	143.4	83.8	76.6	139.6	83.1	74.6	130.2	80.9	43.8	88.8	42.1	87.1	40.6	85.7	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.0	13.0	0.4	98.2	17.2	4.1	-12.0	101.2	-22.0	103.6	-31.6	105.9	-44.1	108.8		
(5)			-0.3	81.3	16.7	1.7	-12.3	82.6	-22.1	83.6	-31.4	84.5	-43.6	85.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	39.7	43.5	49.2	59.2	68.8	75.3	79.0	76.5	71.6	60.3	49.3	43.9
	DBStd	15.77	9.88	9.85	9.91	8.60	7.16	5.44	4.66	4.21	5.37	7.76	8.32	9.34
	HDD50	1095	342	232	143	20	0	0	0	0	0	9	110	239
	HDD65	3478	785	606	496	211	48	2	0	0	12	187	474	657
	CDD50	4664	23	49	118	297	583	759	898	821	649	329	89	49
	CDD65	1573	0	3	6	38	166	311	433	356	211	43	3	3
	CDH74	14098	0	20	81	363	1475	3007	4308	2953	1524	345	12	10
	CDH80	5575	0	2	11	63	503	1252	2025	1156	495	66	0	2
Wind	WSAvg	4.9	5.4	5.8	6.1	6.5	5.0	4.7	4.1	3.4	3.8	4.6	4.8	4.7
Precipitation	PrecAvg	43.3	3.3	2.5	4.2	3.0	3.3	4.1	4.8	4.4	4.4	3.3	3.0	3.3
	PrecMax	56.5	6.1	5.4	8.1	7.4	5.9	10.3	8.1	10.7	18.9	6.8	8.1	6.1
	PrecMin	34.2	1.1	0.5	0.9	1.2	0.9	1.5	2.0	0.6	0.1	0.0	0.6	0.9
	PrecStd	6.3	1.3	1.2	1.7	1.4	1.5	2.4	1.6	2.3	3.8	1.7	1.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.2	77.0	81.5	86.0	91.4	97.2	99.8	95.0	92.6	85.9	75.3	73.2
		MCWB	62.1	64.3	64.0	66.4	73.5	74.4	76.0	76.5	71.8	73.0	63.9	66.3
	2%	DB	64.3	71.8	76.5	81.4	88.5	92.8	96.5	91.1	88.2	81.3	71.9	69.6
		MCWB	57.5	63.4	62.4	66.5	71.9	74.6	77.2	76.2	73.5	69.5	62.7	65.0
	5%	DB	61.2	66.3	71.9	78.8	86.0	90.2	91.5	89.7	85.8	78.5	68.1	63.8
		MCWB	55.4	59.4	59.7	65.0	71.4	75.0	76.9	76.1	72.7	68.1	61.2	59.8
	10%	DB	55.4	62.5	67.8	74.6	81.8	87.8	90.2	86.4	82.1	73.4	63.9	60.5
		MCWB	50.4	55.9	57.4	62.8	69.9	74.0	76.9	75.1	71.8	66.2	57.8	56.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.6	67.4	69.5	72.6	76.4	79.2	82.3	81.4	78.8	77.4	67.6	69.2
		MCDB	69.3	73.2	75.4	79.0	87.5	87.2	90.6	88.5	83.3	83.3	70.8	71.9
	2%	WB	61.2	64.3	65.1	70.0	74.5	77.4	80.8	79.2	76.4	73.7	65.4	64.8
		MCDB	63.0	68.7	73.7	76.7	84.1	87.9	89.6	86.8	83.1	76.7	70.0	67.8
	5%	WB	56.7	61.4	62.7	67.7	73.2	76.4	79.3	77.7	75.2	71.2	62.9	61.4
		MCDB	60.1	65.1	67.9	75.1	82.0	86.6	88.8	84.9	82.4	75.8	66.1	63.5
	10%	WB	51.7	55.4	60.0	65.4	71.8	75.3	77.9	76.2	73.6	68.2	59.6	56.5
		MCDB	55.3	60.7	65.6	72.2	79.4	85.0	87.6	83.8	79.5	72.4	63.9	59.5
Mean Daily Temperature Range	5% DB	MDBR	19.2	20.3	22.1	23.8	21.2	21.2	20.5	18.8	18.9	21.8	21.8	19.3
		MCDBR	25.9	26.5	27.9	25.7	25.0	23.6	22.8	21.3	23.2	24.5	24.8	23.0
		MCWBR	19.2	17.6	15.7	13.2	10.5	8.4	8.3	7.9	9.8	13.5	16.0	17.8
	5% WB	MCDBR	22.7	25.4	22.5	21.2	20.3	20.6	21.0	18.5	18.5	18.2	19.3	20.6
		MCWBR	20.6	19.5	15.5	12.5	9.6	9.1	9.7	8.9	9.6	11.3	15.0	17.8
Clear-Sky Solar Irradiance	taub	0.304	0.319	0.349	0.394	0.442	0.492	0.520	0.518	0.407	0.360	0.322	0.306	
	taud	2.521	2.478	2.427	2.308	2.201	2.110	2.048	2.042	2.350	2.466	2.527	2.555	
	Ebn at Noon	285	292	291	282	269	254	246	244	269	273	275	277	
	Edh at Noon	25	30	34	41	46	50	53	52	36	30	25	23	
All-Sky Solar Radiation	RadAvg	787	1012	1328	1688	1879	2022	1917	1735	1420	1141	869	691	
	RadStd	59	118	102	129	146	129	96	96	121	144	70	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+272	+189

Nomenclature: See separate page